Mth601 Quiz#2
Lecture (23-28)
Due Date (7-9) feb2022

- Mth601-Quiz-2 riz mughal 6feb2021 (pg32)
- Mth601 Quiz#2(lec23 to 28)(7Feb2022) made by stylish (pg72)
- Mth601 Quiz2 feb2022 (pg70)
- MTH601 SS Quiz 2 video23-28+others feb0822 (pg111)
- 3 groups quiz unlimited

Question # 1 of 10 (Start time: 10:05:12 PM, 24 July 2021) Total Marks: 1

In Simplex standard table to solve an LP problem of Minimization, we choose the candidate for entering variable in----------.

Minimization k liyee most positive

Maximization k liyee most negative

Select the correct option

- Constraint with most positive coefficient
- Objective function with most positive coefficient ✓

pg 88

- Constraint with most negative coefficient
- Objective function with most negative coefficient

Click to Save Answer & Move to Next Question

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

While solving a linear programming problem by big M-Method, traditionally the _____ variables are chosen in the initial basic feasible solution.

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- negative slack
- positive slack
- entering
- artificial

(Lecture 23, Pg.94)

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

In the big-M method, if the introduced _____ variables do not leave the basis in the final interaction, then this indicates that the give linear programming problem cant be optimized.

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- entering
- positive slack
- negative slack
- artificial

(Lecture 23, Pg.94)

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Question # 9 of 10

SOLVED BY: WAQAR HUSSAIN

While solving an LP by two phase method, an objective function of 1st phase is always of _____

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- maximization
- minimization
- dependent on original objective function
- None of these

(Lecture 24, Pg.101)

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Question # 3 of 10 (Start time: 03:33:59 PM, 06 February 2021)

If the primal constraint is originally in equation form, the corresponding dual variable is not unrestricted.

Select the correct option

- ○ True
- ◉ False

RIZ MUGHAL

Question # 6 of 10 (Start time: 03:34:45 PM, 06 February 2021) **Total**

While facing degeneracy in solving an LP problem,then the further iterations always assure the non-negative optimal solution.

RIZ MUGHAL

Select the correct option

- ○ True
- ◉ False

MTH601:Quiz

Question # 7 of 10 (Start time: 03:34:59 PM, 06 February 2021)

Dual of a Dual of Dual is------.

Select the correct option

- ○ Dual ✔
- ◉ Primal
- ○ Double dual
- ○ Single dual

RIZ MUGHAL

Question # 1 of 10 (Start time: 05:41:04 PM, 06 February 2021)

Total Mark

If x_1 unrestricted variable and $x_2 = 0$, then standard primal form of objective function of Z. Where Max $Z = 7x_1 + 89x_2$.

Select the correct option

Reload Math Equation

○	Max $Z = 7(x_1^+ + x_1^+) + 89x_2$
○	Max $Z = 7(x_1^+ + x_1^-) + 89x_2$
◉	Max $Z = 7(x_1^+ - x_1^-) + 89x_2$
○	Max $Z = 7(x_1^- + x_1^-) + 89x_2$

RIZ MUGHAL

Question # 2 of 10 (Start time: 05:41:24 PM, 06 February 2021)

Which of the following will be an example of degenerate basic feasible solution for an LP problem?

Select the correct option

- ○ (2,3-1)
- ◉ (0,2,1) ✓
- ○ (2,1,3)
- ○ (-1,2,0)

RIZ MUGHAL

Question # 3 of 10 (Start time: 05:41:40 PM, 06 February 2021) **Total**

While solving a Linear Programming problem by Simplex Method, an inequality constraint is transformed into strict equality by the inclusion of ---------- variable.

Select the correct option

- [x] slack
- [ ] decision
- [ ] artificial
- [ ] basic

RIZ MUGHAL

Question # 5 of 10 (Start time: 05:42:11 PM, 06 February 2021)

The dual of a dual problem yields the original primal.

Select the correct option

- (●) True
- ( ) False

RIZ MUGHAL

Question # 6 of 10 (Start time: 05:42:26 PM, 06 February 2021)

Total Marl

If a variable in the Primal is unrestricted in sign, then the corresponding constraint in the dual will be of -------- type and vice versa.

Select the correct option

- ○ >=
- ○ <=
- ◉ = ✓
- ○ none of these

RIZ MUGHAL

Question # 3 of 10 (Start time: 05:46:27 PM, 06 February 2021)

Degenerate solutions correspond to---------- variables.

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Select the correct option

- Dual
- Non-basic
- Basic ✓
- Unbounded

Question # 10 of 10 (Start time: 05:48:17 PM, 06 February 2021) **Total Marks:**

While solving an LP problem, if the candidate for a leaving variable becomes zero, then solution is said to be --------.

Select the correct option

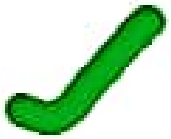
- ○ optimal solution
- ○ infeasible solution
- ○ non-degenerate solution
- ◉ degenerate solution

RIZ MUGHAL

Question # 4 of 10 (Start time: 09:43:41 PM, 09 February 2022)

In Simplex standard table to solve an LP problem of Maximization, we choose the candidate for entering variable in-------.

Select the correct option

○	Objective function with most negative coefficient 
○	Constraint with most positive coefficient
◉	Objective function with most positive coefficient
○	Constraint with most negative coefficient

MTH601 – Operations Research (Quiz No 2) **Quiz Start Time: 10:28 PM, 09 February 2022**

Question # 9 of 10 (Start time: 10:36:27 PM, 09 February 2022) **Total Marks:** 1

When we have $\geq$ type constraints, then we convert it to equality constraints by introducing _____ variable(s) and solve it by _____ method.

Select the correct option

Reload Math Equations

○	artificial, Two phase
○	artificial, Big M
●	artificial and slack, Big M or Two phase ✓
○	slack, Big M

Question # 9 of 10 (Start time: 08:59:32 PM, 09 February 2022) **Total Marks:** 1

In Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column entries should be zero except-----.

Select the correct option

○	key row(pivot)entry
○	row with maximum ratio on RHS column
○	1st row entry
○	last row entry

Question # 5 of 10 (Start time: 10:19:31 PM, 09 February 2022) **Total Marks:** 1

In the Simplex method, if all the elements in columns of the ratio by taking the RHS of each row and dividing by the corresponding element of the key column, are equal, the situation is known as ----------

Select the correct option

○	Tie
○	Unboundedness
○	Degeneracy
○	Non-degeneracy

Question # 2 of 10 (Start time: 10:29:40 PM, 09 February 2022) **Total Marks:** 1

While solving a LP problem by Simplex method in a given iteration, the new basic variable is ---------- and the variable remove from the basis is called ---------.

Select the correct option

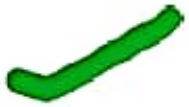
○	(leaving, entering)
○	(infeasible, feasible)
○	(feasible, infeasible)
○	(entering, leaving) ✓

Question # 2 of 10 (Start time: 07:35:04 PM, 09 February 2022)

Total Marks: 1

In M-method, which of the following is true about the coefficient (M) of artificial variable(A) in the objective function?

Select the correct option

- ○ M--› –Infinity
- ○ M--›optimal solution
- ◉ M--›+Infinity 
- ○ M--›zero

Question # 4 of 10 (Start time: 07:36:49 PM, 09 February 2022) **Total Marks:** 1

While solving a Linear Programming problem by Simplex Method, an inequality says:'2x-3y<5' is transformed into strict equality of the form-----, where 's>=0'.

Select the correct option

○	2x-3y −s +A =5
○	2x-3y +s + A=5
○	2x-3y −s =5
○	2x-3y +s =5 ✓

Question # 2 of 10 (Start time: 07:49:56 PM, 09 February 2022) **Total Marks: 1**

In the Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column's entries should be ---------- except key row(pivot) entry.

Select the correct option

- ○ strictly negative
- ○ maximum or minimum
- ○ zero ✓
- ○ strictly positive

MTH601 - Operations Research (Quiz No 2)

Question # 2 of 10 (Start time: 06:58:41 PM, 09 February 2022)

In two phase method, which of the following will be taken as starting solution for 2nd phase?

Select the correct option

○	Optimum feasible solution of 1st phase ✓
○	Non-basic solution of 1st phase containing non-negative slacks
○	Infeasible solution of 1st phase.
○	Basic solution of 1st phase containing non-zero artificial variables

Question # 5 of 10 (Start time: 07:01:39 PM, 09 February 2022) **Total Marks: 1**

While solving a Linear Programming problem by Simplex Method, an inequality says:"2x-3y>5" is transformed into strict equality of the form-----, where 's>=0"

Select the correct option

- ○ 2x-3y -s =5 ✓
- ○ 2x-3y +s + A=5
- ○ 2x-3y -s +A =5
- ○ 2x-3y +s =5

Question # 6 of 10 (Start time: 11:26:05 PM, 09 February 2022) **Total Marks:** 1

In the initial table of Simplex method, the objective function should not contain the terms involving----------variables.

Select the correct option

○	basic
○	non-basic
○	degenerate
○	artificial

Question # 10 of 10 (Start time: 11:34:29 PM, 09 February 2022) **Total Marks: 1**

Once an LP problem is converted to a table, then calculation steps would be same for _____ method(s).

Select the correct option

- (●) all of above
- ( ) Big M
- ( ) simplex
- ( ) Two phase

MTH601 - Operations Research (Quiz No 2)

Question # 1 of 10 (Start time: 10:57:34 PM, 09 February 2022)

A linear programming problem _____ optimal solutions.

Select the correct option

- ○ may have the multiple ✓
- ○ must have the multiple
- ○ must not have the multiple
- ○ may not have the multiple

Question # 2 of 10 (Start time: 10:57:57 PM, 09 February 2022) **Total Marks: 1**

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '5' variables. Then how many Combinations of Basic feasible Solutions may possible?

Select the correct option

- ○ 2
- ○ 24
- ○ 10
- ○ 15 ✓

MTH601 - Operations Research (Quiz No 2)

Question # 6 of 10 (Start time: 11:11:07 PM, 09 February 2022)

The cost coefficient of artificial variable in Objective function is ----------.

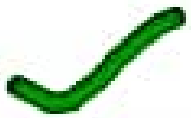
Select the correct option

○	0
○	> than 1
○	1
○	M ✓

Question # 1 of 10 (Start time: 10:49:30 PM, 09 February 2022)

In an LP problem, to evaluate the basic variables in Simplex method, which of the following method is used?

Select the correct option

- Gauss Elimination method
- Gauss-Jordan method 
- Gauss Least Square method
- Gauss-Seidel method

MTH601 - Operations Research (Quiz No 2)

Quiz Start Time: 10:52 PM, 09 February 2022

Question # 1 of 10 (Start time: 10:52:29 PM, 09 February 2022)

Total Marks: 1

In the Simplex method to solve an LP problem of minimization, if at the end of iteration, every entry of objective function row is negative then the given problem ----------.

Select the correct option

- has no solution
- needs further improvement
- has been minimized
- can not be optimized

Question # 5 of 10 (Start time: 10:01:39 PM, 09 February 2022)

In M-method, which of the following is the standard equality corresponds to the constraint:"3x+5y=2"?

Select the correct option

○	3x+2y−A=2
◉	3x+2y−MA=2
○	3x+2y +A=2
○	3x+2y+MA=2

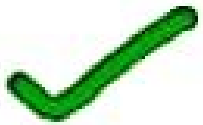
Not confirm

Question # 6 of 10 (Start time: 11:30:41 PM, 09 February 2022)

Total Marks: 1

In the Simplex method to solve an LP problem of maximization, if at the end of iteration, every entry of objective function row is positive then the given problem----------.

Select the correct option

- ○ can not be optimized
- ◉ has been maximized 
- ○ has no solution
- ○ needs further improvement

Question # 3 of 10 (Start time: 11:03:04 PM, 09 February 2022) **Total Marks: 1**

In the Simplex method to solve an LP problem of minimization, if at the end of iteration, there is a positive coefficient in the objective row then the given problem ----------.

Select the correct option

- ○ can not be optimized
- ○ needs further improvement
- ◉ has been maximized
- ○ has no solution

Question # 2 of 10 (Start time: 10:01:06 PM, 09 February 2022) **Total Marks: 1**

While solving an LP problem by Simplex method, the inclusion of slacks in the constraints' inequalities helps in finding ---------- variables.

Select the correct option

- (●) basic
- ( ) non-basic
- ( ) decision
- ( ) artificial

Question # 5 of 10 (Start time: 09:27:53 PM, 09 February 2022) **Total Marks: 1**

In an LP problem, to evaluate the basic variables in Simplex method in any iteration, we would have to express Objective function in terms of ---------- variables.

Select the correct option

- Slack
- Artificial
- Non-basic
- Basic

Question # 3 of 10 (Start time: 10:01:21 PM, 09 February 2022)

Total Marks: 1

If the original Linear Programming problem has no Feasible Solution, then either the Big-M method or Phase-I of the Two-Phase method will give a final solution that has at least one artificial variable _____.

Select the correct option

- less than zero
- of any arbitrary value
- greater than zero ✓
- equal to zero

Question # 9 of 10 (start time: 09:52:13 PM, 09 February 2022) **Total Marks: 1**

By Simplex method, to minimize 'z = 8x−2y' of an LP problem, if 'z=A>0' for the initial iteration then for its next improved solution(0<A<100), which of the following would be the next entering variable?

Select the correct option

- ○ x>0
- ◉ y>0 ✓
- ○ y<0
- ○ x<0

Question # 10 of 10 (start time: 09:52:20 PM, 09 February 2022)

Total Marks: 1

In M-method, if the given LP problem has the feasible solution then the optimization algorithm enforces artificial variable to--------.

Select the correct option

- (●) Zero
- ( ) + Infinity
- ( ) Optimal solution
- ( ) – Infinity

MTH601 - Operations Research (Quiz No 2)

Question # 1 of 10 (Start time: 10:00:55 PM, 09 February 2022)

An optimal solution of a Linear Programming problem is necessarily a Basic feasible solution.

Select the correct option

○	False
◉	True 

Question # 3 of 10 (Start time: 09:50:58 PM, 09 February 2022)

For $Z = 4x_1 + x_2$ subject to $3x_1 + 4x_2 \geq 20, x_1 + 5x_2 \geq 15$, we have _____ variables.

Select the correct option

◉	two slack
○	three objective
○	six
○	one artificial

Question # 6 of 10 (Start time: 09:51:36 PM, 09 February 2022)

Which of the following is true about an LP problem which is solved by M-method and having infeasible solution?

Select the correct option

○	Associated constraints are not consistent
◉	All choices are equivalent. 
○	Final iteration table include one of the artificial variables.
○	Penalty 'M' may not able to force artificial variable to zero in final iteration.

Question # 8 of 10 (Start time: 09:52:03 PM, 09 February 2022)

Total Marks: 1

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, then in a given iteration, how many Basic feasible VARIABLES are possible to exists?

Select the correct option

- (●) 4 
- ( ) 6
- ( ) 2
- ( ) 10

MTH601 - Operations Research (Quiz No 2)

Question # 10 of 10 (Start time: 09:45:20 PM, 09 February 2022)

Shortcoming of Big M method is that the value of M could be _____.

Select the correct option

○	negative
○	very small
○	positive
◉	very large

Question # 1 of 10 (Start time: 09:50:28 PM, 09 February 2022)

Total Marks: 1

In Simplex Standard table to solve an LP problem, if the ratio by taking the RHS of each row and dividing by the corresponding element of the key column is {2/3,5,1/3, -1/2}, then which of the following variable will be leaving?

Select the correct option

- ○ Variable corresponding to '-1/2'
- ○ Variable corresponding to '2/3'
- ◉ Variable corresponding to '1/3' ✓
- ○ Variable corresponding to '5'

Question # 2 of 10 (Start time: 09:50:36 PM, 09 February 2022)

In two phase method process, first phase _____ the sum of artificial variables.

Select the correct option

◉	minimize 
○	non of these
○	maximize or minimize depending on the situation
○	maximize

Question # 7 of 10 (Start time: 09:44:32 PM, 09 February 2022) **Total Marks: 1**

In the final iteration of M-method to obtain the optimal solution, the artificial variable must------- the basis.

Select the correct option

- subtract from each variable of
- add in each variable of
- include in
- leave ✓

Question # 9 of 10 (Start time: 09:45:10 PM, 09 February 2022)

When constraints are of greater than or equal to type, then we _____.

Select the correct option

- ○ both of these
- ○ add slack variable and subtract artificial varialbe
- ◉ subtract slack variable and add artificial variable 
- ○ non of these

Question # 10 of 10 (Start time: 09:29:50 PM, 09 February 2022)

The dual of the dual of a Linear Programming problem is itself the _____ problem.

Select the correct option

○	dual
◉	primal
○	transportation
○	assignment

MTH601 - Operations Research (Quiz No 2)

Question # 3 of 10 (Start time: 09:43:31 PM, 09 February 2022)

To remove artificial variable(s) from the objective function, we have _____

Select the correct option

○	simplex
◉	both b and c
○	Big M
○	Two Phase

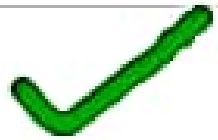
sec(s)

MTH601 - Operations Research (Quiz No 2) **Quiz Start Time: 09:26 PM, 09 February 2022**

Question # 7 of 10 (Start time: 09:29:12 PM, 09 February 2022) **Total Marks: 1**

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, if the initial basic feasible solution is say;(2,4,0,2), then it is ---------- solution.

Select the correct option

- non-degenerate feasible
- degenerate feasible 
- non-degenerate infeasible
- degenerate infeasible

Question # 8 of 10 (Start time: 09:29:22 PM, 09 February 2022)

Which of the following difficulty may found while attempting an LP problem by M-method?

Select the correct option

○	Degeneracy is inevitable
○	Artificial do not leave the basis
◉	Computational error due to large value of M 
○	It often leads to infeasible solution

MTH601 - Operations Research (Quiz No 2)

Question # 9 of 10 (Start time: 09:29:34 PM, 09 February 2022)

The positive slack variables indicate ----------.

Select the correct option

○	nil resources
○	excess resource available 
○	idle(waste) resource
◉	shortage of resources

Question # 3 of 10 (Start time: 09:27:26 PM, 09 February 2022)

Under which of the following condition to solve an LP by using two phase method, we can't proceed for 2nd phase?

Select the correct option

- (●) Objective function of 1st phase has positive value. 
- ( ) Objective function of 1st phase has zero value.

Question # 4 of 10 (Start time: 09:27:36 PM, 09 February 2022)

If in a LP problem, the objective is to maximize: z = 2x+3y, with all associated constraints of '=<' type, then in the 1st iteration, z = -

Select the correct option

○	1
○	6
○	5
◉	0

Question # 6 of 10 (Start time: 09:28:45 PM, 09 February 2022) **Total Marks: 1**

In two phase method, for the phase-I, if the objective function has zero value with all vanishing artificial variables then we -------.

Select the correct option

- ○ Infeasible solution
- ○ cant proceed for 2nd phase
- ○ necessarily have optimal solution
- ◉ proceed for 2nd phase 

Question # 10 of 10 (Start time: 11:31:43 PM, 09 February 2022) **Total Marks: 1**

By Simplex method, to maximize 'z = 2x−9y'of an LP problem, if 'z=0' for the initial iteration then for its next improved solution, which of the following would be the next entering variable?

Select the correct option

- ○ x<0
- ○ y<0
- ○ y>0
- ◉ x>0 ✓

MTH601 - Operations Research (Quiz No 2)

Question # 1 of 10 (Start time: 09:26:49 PM, 09 February 2022)

In two phase method, for the phase-I, if the given problem has feasible solution then----

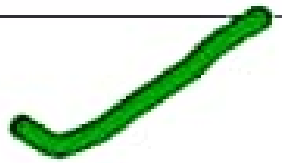
Select the correct option

○	objective function is zero but artificial may arbitrary
○	objective function is arbitrary but artificial is zero
◉	both objective function and artificial are zero ✔
○	both objective and artificial can have arbitrary values

Question # 2 of 10 (Start time: 09:27:16 PM, 09 February 2022) **Total Marks: 1**

In two phase method if the minimum value of objective function in the first phase is greater than zero, then the solution of original problem _____.

Select the correct option

- ○ non of these
- ○ is uncertain
- ○ exists
- ◉ does not exists 

Question # 9 of 10 (Start time: 11:31:30 PM, 09 February 2022)

An unrestricted variable can always be expressed as the difference of two _____ variables.

Select the correct option

○	none of these
◉	non-negative 
○	positive
○	negative

Question # 1 of 10 (Start time: 11:29:46 PM, 09 February 2022) **Total Marks: 1**

While solving an LP problem by the Simplex method, in the standard table, the element at the intersection of key column and key row is called ---------- element.

Select the correct option

- Entering
- Pivot
- Slack
- Leaving

MTH601 - Operations Research (Quiz No 2)

Question # 5 of 10 (Start time: 11:27:28 PM, 09 February 2022)

The inequality $3x - 2y \geq 14$ is equivalent to_____.

Select the correct option

○	$3x - 2y \leq 14$
○	$-3x + 2y > 14$
○	$3x - 2y < -14$
◉	$-3x + 2y \leq -14$ 

MTH601 - Operations Research (Quiz No 2)

Question # 6 of 10 (Start time: 11:27:51 PM, 09 February 2022)

In two phase method, for the phase-I, a new objective function is expressed as -----.

Select the correct option

○	sum of slack variables
○	difference of slack variables
○	difference of artificial variables
◉	sum of artificial variables 

Question # 7 of 10 (Start time: 11:14:18 PM, 09 February 2022) **Total Marks: 1**

The insensitivity of the solution relative to the original decision variables in an LP problem which is solved by M-method is overcome by ---------.

Select the correct option

- ○ Duality principle
- ○ Graphical method
- ◉ Two phase method 
- ○ Simplex method

Question # 10 of 10 (Start time: 11:14:47 PM, 09 February 2022) Total Marks: 1

In a feasible region, if the end points of the line segment are basic solutions, then all the points between these two will ----------.

Select the correct option

- be non basic
- be optimal
- be infeasible
- also be the basic

Question # 2 of 10 (Start time: 11:17:19 PM, 09 February 2022) **Total Marks: 1**

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables. If the initial basic feasible solution is say;(2,4,1,2), then its corresponding non-basic solution is---------.

Select the correct option

- ○ (2,2)
- ○ (1,2)
- ○ (-4,-2)
- ◉ (0,0) ✓

Question # 6 of 10 (Start time: 11:07:47 PM, 09 February 2022) **Total Marks: 1**

If an artificial variable is not included while converting a constraint of type ">=" into equation, then we will have -------- solution.

Select the correct option

- degenerate
- non-degenerate
- feasible
- infeasible 

Question # 9 of 10 (Start time: 11:08:45 PM, 09 February 2022) **Total Marks: 1**

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, then in a given iteration, how many infeasible VARIABLES are possible to exists?

Select the correct option

- (●) 10
- ( ) 2
- ( ) 4
- ( ) 6



4 or
Not confirm

MTH601 - Operations Research (Quiz No 2)

Question # 1 of 10 (Start time: 11:12:43 PM, 09 February 2022)

When using Big M method, the objective function should not contain _____ variable(s).

Select the correct option

○	artificial and slack
◉	artificial
○	x_1
○	slack

Question # 2 of 10 (Start time: 11:06:47 PM, 09 February 2022) Total Marks: 1

In Big M-method, if the objective function is expressed in terms of non-basic variable then the coefficient of artificial variable in this iteration must be --------.

Select the correct option

- M
- zero 
- -M
- one

Question # 3 of 10 (Start time: 11:06:57 PM, 09 February 2022) **Total Marks: 1**

In two phase method, for the phase-I, the problem has infeasible solution if the minimum value of objective function ----------.

Select the correct option

- (●) greater than zero
- ( ) less than zero
- ( ) is imaginary
- ( ) equal to zero

Question # 5 of 10 (Start time: 11:07:34 PM, 09 February 2022) **Total Marks: 1**

If initial basic solution is ----------, while solving an LP problem then no further iteration can be performed.

Select the correct option

- (●) infeasible
- ( ) degenerate
- ( ) feasible
- ( ) non-degenerate

MTH601 - Operations Research (Quiz No 2) Quiz Start Time: 11:06 PM, 09 February 2022

Question # 1 of 10 (Start time: 11:06:39 PM, 09 February 2022) Total Marks: 1

Which of the following order pair would minimize the objective function of the linear programming problem; $z = x + 5y$ subject to $x \geq 2, y \geq 0$?

Select the correct option

Reload Math Equations

○	(2,0)
○	(2,3)
◉	(0,0)
○	(0,3)

Question # 2 of 10 (Start time: 11:02:47 PM, 09 February 2022)

In Simplex method, which of the following is the standard equality corresponding to the constraint"3x+5y>=2"?

Select the correct option

◉	3x+2y−S+A=2 
○	3x+2y+S+A=2
○	3x+2y−S−A=2
○	3x+2y+S−A=2

Question # 1 of 10 (Start time: 11:02:35 PM, 09 February 2022)

For $Z = 4x_1 + x_2$ subject to $3x_1 + 4x_2 \geq 20, x_1 + 5x_2 \geq 15$, we have _____ variables.

Select the correct option

○	one slack
◉	two artificial 
○	six
○	three objective

Question # 10 of 10 (start time: 10:56:14 PM, 09 February 2022) **Total Marks: 1**

Which of the following is the standard form of objective function corresponding to,"Minz=2x−11y, subject to x=5 and y=7"? Where A's are artificial.

Select the correct option

- z = 2x − 11y −MA1 − MA2
- z = 2x − 11y +MA1 − MA2
- z = 2x − 11y +MA1 + MA2 ✓
- z = 2x − 11y −MA1 + MA2

Question # 7 of 10 (Start time: 10:09:34 PM, 09 February 2022) **Total Marks: 1**

In the Simplex method to solve an LP problem of maximization, if at the end of an iteration, there is a negative coefficient in the objective row then the given problem---------.

Select the correct option

- ○ has been maximized
- ◉ needs further improvement 
- ○ cant be optimized
- ○ has no solution

Question # 6 of 10 (Start time: 10:55:13 PM, 09 February 2022)

For $Z = 4x_1 + x_2$ subject to $3x_1 + 4x_2 \geq 20, x_1 + 5x_2 \geq 15$, we have _____ variables.

Select the correct option

○	three objective
◉	both a and b
○	two slack
○	two artificial

Question # 9 of 10 (Start time: 10:56:05 PM, 09 February 2022) **Total Marks: 1**

If any or all of the artificial variables do not leave the basis in the final solution, then this indicates that the problem _____ .

Select the correct option

- ○ may have a solution
- ○ non of these
- ○ must have a solution
- ◉ does not have a solution 

Question # 7 of 10 (Start time: 10:05:36 PM, 09 February 2022)

In two phase method, for the phase-I, a new objective function in terms of artificial is to be ----------.

Select the correct option

○	maximized
◉	minimized

MTH601 - Operations Research (Quiz No 2)

Question # 8 of 10 (Start time: 10:05:52 PM, 09 February 2022)

A Linear Programming problem may not have any feasible solution.

Select the correct option

◉	True 
○	False

Question # 9 of 10 (Start time: 10:06:05 PM, 09 February 2022)

Which of the following is true about the inclusion of non-negative slack variable into a constraint of type 'less than or equal'?

Select the correct option

○	This inclusion doest not affect the inequality
○	Inequality become redundant and hence skips from the LP problem.
○	Inequality is transformed into type: 'greater than or equal'
◉	Inequality is transformed into strict equality constraint 

Question # 6 of 10 (Start time: 10:09:25 PM, 09 February 2022)

Total Marks: 1

In two phase method if the minimum value of objective function in the first phase is zero, then the solution of original problem _____.

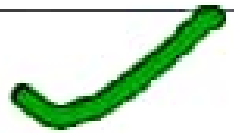
Select the correct option

- non of these
- is uncertain
- does not exists
- exists 

Question # 2 of 10 (Start time: 10:04:36 PM, 09 February 2022) **Total Marks: 1**

While solving an LP problem by Simplex method, if we have '3' constraint equations in '5' variables then to find basic variables we have to put any ------- variables equal to zero.

Select the correct option

- ○ 4
- ○ 2
- ◉ 3 
- ○ 1

MTH601 - Operations Research (Quiz No 2)

Question # 3 of 10 (Start time: 10:04:51 PM, 09 February 2022)

In graphical solution of solving Linear Programming problem to convert inequalities into equations, we ----------.

Select the correct option

○	use Artificial surplus variables
◉	simply assume them to be equations 
○	use Surplus variables
○	use Slack variables

MTH601 - Operations Research (Quiz No 2)

Question # 4 of 10 (Start time: 10:05:01 PM, 09 February 2022)

In the initial iteration of Big M-method, the artificial variables appear in ---------.

Select the correct option

◉	Basis 
○	Non-basic variables' set

Time Left sec(s)

MTH601 - Operations Research (Quiz No 2)

Quiz Start Time: 10:00 PM, 09 February 2022

Question # 7 of 10 (Start time: 10:02:17 PM, 09 February 2022)

Total Marks: 1

In a feasible region, if the end points of a line segment are basic solutions which BOTH also give the optimal solution, then ---------- will also serve for optimal.

Select the correct option

- (●) all other points on the segment
- ( ) the point which divides segment in ratio 1:2
- ( ) no other point
- ( ) the only midpoint of the segment